

Global Pollution Analysis and Prediction Using Machine Learning

1. Project Title

Global Pollution Analysis and Prediction Using Machine Learning

2. Dataset Information

- Dataset Name: Global Pollution Dataset 2025–2026
 - Dataset Source: Kaggle
 - Dataset Link: <https://www.kaggle.com/datasets/atifkhan12/global-air-pollution-dataset-2025-2026>
 - Number of Rows: 17,472
 - Number of Columns: 12
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3. Target Variable

The target variable used in this project is:

AQI_Class

The AQI Class column represents air quality categories such as healthy or unhealthy air conditions.

4. Type of Task

This project is a:

Classification Task because the target variable contains categorical values.

5. Problem Statement

Air pollution is one of the major environmental challenges affecting human health and climate worldwide. Predicting air quality levels can help governments and environmental agencies take preventive measures and improve public awareness. The purpose of this project is to analyze pollution-related environmental factors and predict AQI classification using supervised machine learning algorithms.

6. Research Questions

1. Which environmental factors most affect AQI classification?
2. Can machine learning models accurately classify air quality levels?

3. Which machine learning algorithm performs best for AQI prediction?
 4. How do PM2.5 and PM10 levels impact air quality?
 5. What is the relationship between gases such as NO2, SO2, CO, and AQI classification?
 6. Does preprocessing improve machine learning performance?
 7. Which features are most important in predicting AQI classification?
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7. Proposed Methodology

The following methodology will be used in this project:

1. Load and explore the dataset
 2. Perform data preprocessing
 3. Handle missing values
 4. Encode categorical variables
 5. Split the dataset into training and testing sets
 6. Train supervised machine learning models
 7. Evaluate model performance using classification metrics
 8. Compare model results using tables and visualizations
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8. Machine Learning Models

The following machine learning algorithms are used:

- Logistic Regression
 - Decision Tree Classifier
 - Random Forest Classifier
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9. Evaluation Metrics

The following evaluation metrics are used to measure model performance:

- Accuracy Score
- Confusion Matrix
- Classification Report
- Precision
- Recall

- F1-Score
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10. Expected Figures and Tables

Expected Figures

- Model Accuracy Comparison Graph
- Confusion Matrix Heatmap

Expected Tables

- Dataset Summary Table
 - Missing Values Table
 - Machine Learning Model Comparison Table
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11. Expected Outcome

The expected outcome of this project is to identify the most accurate machine learning model for AQI classification and understand the environmental factors contributing to air pollution levels.